

| No.                                                                                                                                                 | Time                       | Source         | Destination    | Protocol |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------|----------------|----------|
| Length Info                                                                                                                                         |                            |                |                |          |
| 780                                                                                                                                                 | 2025-10-05 13:12:52.196682 | 172.16.0.116   | 128.119.245.12 | HTTP 663 |
| GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1                                                                                             |                            |                |                |          |
| Frame 780: 663 bytes on wire (5304 bits), 663 bytes captured (5304 bits) on interface en0, id 0                                                     |                            |                |                |          |
| Ethernet II, Src: b2:6f:a9:ed:a9:d8 (b2:6f:a9:ed:a9:d8), Dst: Routerboardc_67:31:81 (cc:2d:e0:67:31:81)                                             |                            |                |                |          |
| Internet Protocol Version 4, Src: 172.16.0.116, Dst: 128.119.245.12                                                                                 |                            |                |                |          |
| Transmission Control Protocol, Src Port: 49716, Dst Port: 80, Seq: 1, Ack: 1, Len: 597                                                              |                            |                |                |          |
| Source Port: 49716                                                                                                                                  |                            |                |                |          |
| Destination Port: 80                                                                                                                                |                            |                |                |          |
| [Stream index: 29]                                                                                                                                  |                            |                |                |          |
| [Stream Packet Number: 4]                                                                                                                           |                            |                |                |          |
| [Conversation completeness: Complete, WITH_DATA (31)]                                                                                               |                            |                |                |          |
| [TCP Segment Len: 597]                                                                                                                              |                            |                |                |          |
| Sequence Number: 1 (relative sequence number)                                                                                                       |                            |                |                |          |
| Sequence Number (raw): 1944316952                                                                                                                   |                            |                |                |          |
| [Next Sequence Number: 598 (relative sequence number)]                                                                                              |                            |                |                |          |
| Acknowledgment Number: 1 (relative ack number)                                                                                                      |                            |                |                |          |
| Acknowledgment number (raw): 153200505                                                                                                              |                            |                |                |          |
| 1000 .... = Header Length: 32 bytes (8)                                                                                                             |                            |                |                |          |
| Flags: 0x018 (PSH, ACK)                                                                                                                             |                            |                |                |          |
| Window: 2053                                                                                                                                        |                            |                |                |          |
| [Calculated window size: 131392]                                                                                                                    |                            |                |                |          |
| [Window size scaling factor: 64]                                                                                                                    |                            |                |                |          |
| Checksum: 0xcf95 [unverified]                                                                                                                       |                            |                |                |          |
| [Checksum Status: Unverified]                                                                                                                       |                            |                |                |          |
| Urgent Pointer: 0                                                                                                                                   |                            |                |                |          |
| Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps                                                                             |                            |                |                |          |
| [Timestamps]                                                                                                                                        |                            |                |                |          |
| [SEQ/ACK analysis]                                                                                                                                  |                            |                |                |          |
| TCP payload (597 bytes)                                                                                                                             |                            |                |                |          |
| Hypertext Transfer Protocol                                                                                                                         |                            |                |                |          |
| GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n                                                                                         |                            |                |                |          |
| Request Method: GET                                                                                                                                 |                            |                |                |          |
| Request URI: /wireshark-labs/INTRO-wireshark-file1.html                                                                                             |                            |                |                |          |
| Request Version: HTTP/1.1                                                                                                                           |                            |                |                |          |
| Host: gaia.cs.umass.edu\r\n                                                                                                                         |                            |                |                |          |
| Connection: keep-alive\r\n                                                                                                                          |                            |                |                |          |
| Upgrade-Insecure-Requests: 1\r\n                                                                                                                    |                            |                |                |          |
| User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36\r\n               |                            |                |                |          |
| Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n |                            |                |                |          |
| Accept-Encoding: gzip, deflate\r\n                                                                                                                  |                            |                |                |          |
| Accept-Language: en-US,en-GB;q=0.9,en;q=0.8,vi-VN;q=0.7,vi;q=0.6\r\n                                                                                |                            |                |                |          |
| If-None-Match: "51-64026ae9a23b1"\r\n                                                                                                               |                            |                |                |          |
| If-Modified-Since: Thu, 02 Oct 2025 05:59:02 GMT\r\n                                                                                                |                            |                |                |          |
| \r\n                                                                                                                                                |                            |                |                |          |
| [Response in frame: 786]                                                                                                                            |                            |                |                |          |
| [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]                                                              |                            |                |                |          |
| No.                                                                                                                                                 | Time                       | Source         | Destination    | Protocol |
| Length Info                                                                                                                                         |                            |                |                |          |
| 786                                                                                                                                                 | 2025-10-05 13:12:52.463139 | 128.119.245.12 | 172.16.0.116   | HTTP 504 |
| HTTP/1.1 200 OK (text/html)                                                                                                                         |                            |                |                |          |
| Frame 786: 504 bytes on wire (4032 bits), 504 bytes captured (4032 bits) on interface en0, id 0                                                     |                            |                |                |          |
| Ethernet II, Src: Routerboardc_67:31:81 (cc:2d:e0:67:31:81), Dst: b2:6f:a9:ed:a9:d8 (b2:6f:a9:ed:a9:d8)                                             |                            |                |                |          |
| Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.16.0.116                                                                                 |                            |                |                |          |
| Transmission Control Protocol, Src Port: 80, Dst Port: 49716, Seq: 1, Ack: 598, Len: 438                                                            |                            |                |                |          |
| Source Port: 80                                                                                                                                     |                            |                |                |          |
| Destination Port: 49716                                                                                                                             |                            |                |                |          |
| [Stream index: 29]                                                                                                                                  |                            |                |                |          |
| [Stream Packet Number: 6]                                                                                                                           |                            |                |                |          |
| [Conversation completeness: Complete, WITH_DATA (31)]                                                                                               |                            |                |                |          |
| [TCP Segment Len: 438]                                                                                                                              |                            |                |                |          |

```
Sequence Number: 1      (relative sequence number)
Sequence Number (raw): 153200505
[Next Sequence Number: 439      (relative sequence number)]
Acknowledgment Number: 598      (relative ack number)
Acknowledgment number (raw): 1944317549
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Window: 236
[Calculated window size: 30208]
[Window size scaling factor: 128]
Checksum: 0xdf3d [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]
TCP payload (438 bytes)
Hypertext Transfer Protocol
HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Date: Sun, 05 Oct 2025 06:12:52 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/
v5.16.3\r\n
    Last-Modified: Sun, 05 Oct 2025 05:59:01 GMT\r\n
    ETag: "51-6406308156f5d"\r\n
    Accept-Ranges: bytes\r\n
    Content-Length: 81\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=UTF-8\r\n
\r\n
    [Request in frame: 780]
    [Time since request: 0.266457000 seconds]
    [Request URI: /wireshark-labs/INTRO-wireshark-file1.html]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]
    File Data: 81 bytes
Line-based text data: text/html (3 lines)
```